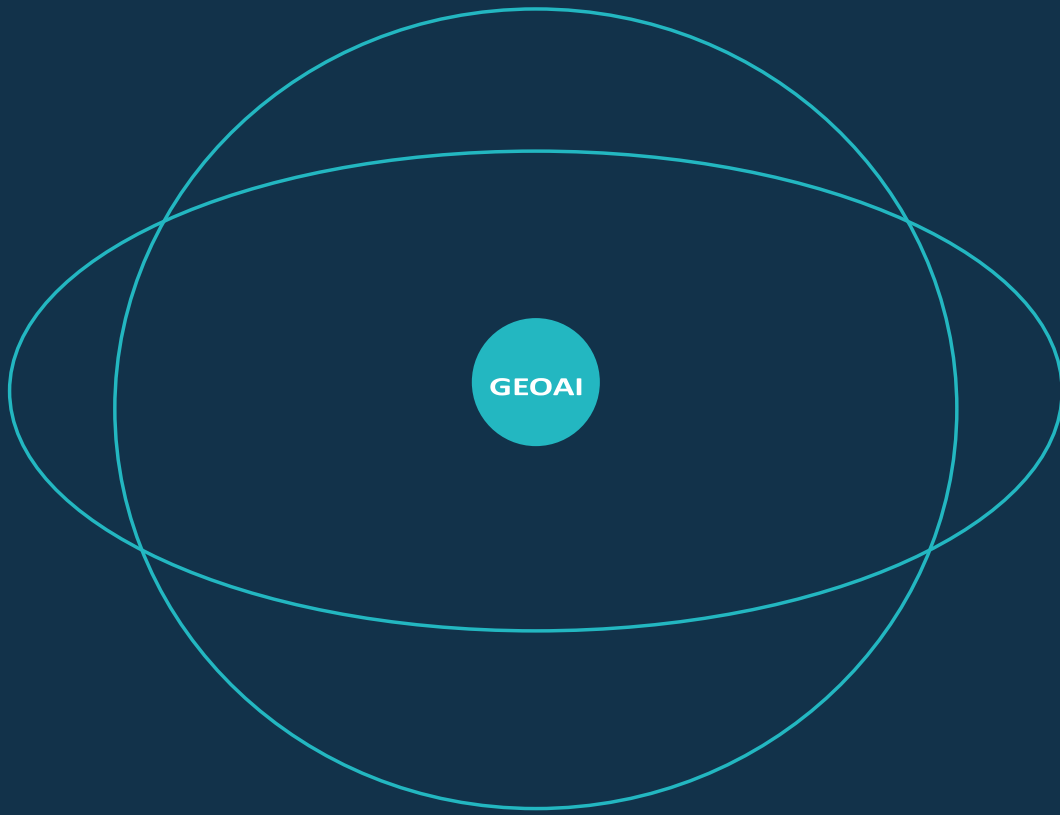




GeoAI: Satellite Data Processing Land Classification System

Comprehensive Technical Project Report

Satellite Processing • Spatial AI • Interactive Maps

DOCUMENT CONTENTS

Six-section technical report with clickable navigation and visual implementation evidence placeholders.

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Project Goal	Process satellite data to calculate NDVI and NDBI, perform land classification, validate outputs, and visualize results interactively.
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INTERACTIVE DOCUMENT FEATURES

Section entries above are clickable bookmarks. Page-level navigation and the visual workflow are designed for screen viewing as well as print/PDF conversion.

1. EXECUTIVE SUMMARY

GeoAI: Satellite Data Processing & Land Classification System is a production-oriented geospatial AI workflow that converts satellite-derived information into actionable spatial insights. The system bridges raw imagery and decision-ready outputs through a controlled sequence of preprocessing, spectral-index computation, land classification, validation, error analysis, and interactive mapping.

The analytical core is centered on two normalized geospatial indices: NDVI for vegetation analysis and NDBI for built-up area analysis. These indicators transform raw spectral measurements into interpretable spatial features that can support the land-classification stage.

Google Colab provides the computational environment for data processing and model training, while Python and Pandas support data preparation, validation, and analysis. The resulting outputs are surfaced through a Streamlit application that enables interactive exploration of maps, classification results, and tabular evidence.

The project also includes a dedicated validation layer using 50+ manual verification points and an Error_Analysis.csv file containing 15+ documented error cases. This evidence-oriented design makes the system traceable and provides a structured basis for diagnosing spatial discrepancies.

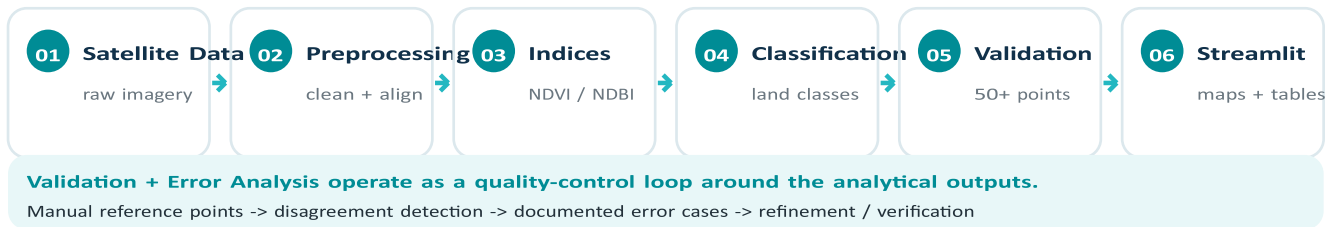
Core Technology Stack

Component	Technology	Responsibility
Data Processing & Model Training	Google Colab	Satellite-data processing, feature preparation, analysis, and model training
Programming Language	Python	Numerical processing, orchestration, application logic, and geospatial workflow implementation
Data Analysis	Pandas	CSV ingestion, validation analysis, row-level checks, and tabular processing
Geospatial Processing	Python geospatial libraries	Coordinate handling, spatial data preparation, mapping support, and geospatial calculations
Geospatial Indices	NDVI / NDBI	Vegetation and built-up area feature generation
Interactive Application	Streamlit	Interactive dashboard interface, stateful controls, map display, and output presentation
Validation & Error Analysis	CSV artifacts	Manual reference points and documented error cases for verification and troubleshooting

System value: The architecture creates a reproducible bridge from raw satellite measurements to interpretable land-classification evidence while keeping validation and error review visible to the user.

2. SYSTEM ARCHITECTURE & WORKFLOW

The system is organized as four modular processing stages surrounded by a validation and error-analysis layer. Each module has a clear contract: prepare data, compute informative spatial features, classify the land surface, then present and verify the resulting outputs.

**MODULE 1****Data Ingestion & Preprocessing**

Load satellite-derived data, inspect structure, handle missing values, preserve spatial context, and prepare spectral inputs.

MODULE 2**Geospatial Index Calculation**

Derive NDVI for vegetation characteristics and NDBI for built-up / urban characteristics using normalized spectral relationships.

MODULE 3**Land Classification Model**

Transform prepared spatial features and index information into interpretable land classes for downstream validation.

MODULE 4**Interactive Visualizations & Mapping**

Expose model outputs through Streamlit maps, indicators, tables, and interactive spatial exploration.

Validation Layer

Manual validation points are maintained as structured CSV records and associated with expected land characteristics. The validation layer checks whether system-generated classifications are spatially consistent with the reference evidence. Mismatches are routed into the error-analysis workflow rather than being silently discarded.

Error Analysis Process

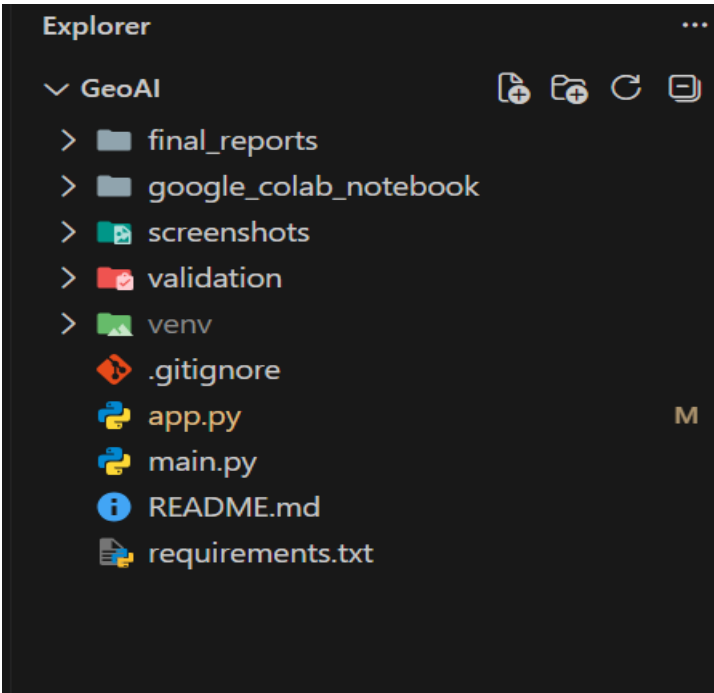
The Error_Analysis.csv artifact captures 15+ real-world error cases. Each case is treated as a troubleshooting record: identify the visible symptom, isolate the root cause, apply a resolution, and record the result. This produces a repeatable engineering feedback loop for spatial quality assurance.

End-to-end operational flow: Satellite imagery → data validation → preprocessing → spectral feature preparation → NDVI/NDBI computation → land classification → manual validation → error analysis → interactive Streamlit visualization.

3. TECHNICAL IMPLEMENTATION & DIRECTORYSTRUCTURE

The repository separates computational work, application logic, validation evidence, and visual assets so the system can be processed in Colab and consumed through the Streamlit interface without mixing core artifacts.

➤ Exact repository layout



Implementation Responsibilities

Artifact	Technical Responsibility
GeoAI.ipynb	Primary Colab notebook for satellite-data ingestion, preprocessing, NDVI/NDBI calculations, analytical checks, classification workflow, and model-training steps.
app.py	Streamlit interface layer responsible for dashboard controls, result presentation, interactive tables, and map visualization.
main.py	Application/bootstrap integration entry point supporting execution flow, shared orchestration, or runtime setup around the dashboard.
Validation_Points.csv	Manual validation evidence containing 50+ reference points used to compare generated classifications with observed/verified land characteristics.
Error_Analysis.csv	Structured troubleshooting evidence containing 15+ documented errors, causes, resolutions, and observed outcomes.
screenshots/	Evidence folder for dashboard, map, and hotspot-response outputs used in technical documentation and presentation.

Output and Evidence Artifacts

The report reserves dedicated evidence panels for the three required outputs. The labels below preserve the exact screenshot paths specified for the project package.

- Streamlit Dashboard View Frontend

Deploy

Analysis Filters

Select District

Multan

Selected Year

2026

GeoGuardian: Urban Expansion Monitor

Build By Muhammad Khurram - AI/ML Intern

Monitoring agricultural land loss and urban builtup expansion using Sentinel-2 and Rnandom Forest

Statistics for Multan (2026)

Total AOI

698.41 sq km

Agricuture Lost

51.3 sq km

Percentage Converted

7.35%

Built-up Area

238.23 sq km

Multan Micro Lvel Zonel Analysis (Top Expansion Hotspots)

Area Name/ Location	Converted Area (sq km)	Total Impact (%)
منطقہ گلشن, Gulshan, ملتان	3.31	0.47
تھریل ملتان مندرجہ منطع ملتان, Basti Bahawal Khera, ملتان	2.76	0.40
منطقہ گلشن, Gulshan, ملتان	2.21	0.32

Download Full data (csv)

Download Boundary & Table Data (GeoJSON)

AI Generated Insight & Policy Recommendations

CRITICAL ALERT: In 2026, Multan's built-up area is calculated at 238.23 sq km using real satellite pixels. Exactly 51.3 sq km of agricultural land was confirmed as converted to built-up infrastructure since 2016.

➤ Stats Multan (2025) View

Deploy

Analysis Filters

Select District

Multan

Selected Year

2025

Build By Muhammad Khurram - AI/ML Intern

Monitoring agricultural land loss and urban builtup expansion using Sentinel-2 and Rnandom Forest

Statistics for Multan (2025)

Total AOI

533.73 sq km

Agricuture Lost

24.22 sq km

Percentage Converted

4.54%

Built-up Area

206.25 sq km

Interactive Map

➤ Map Visualization View

Analysis Filters

Select District

Multan

Selected Year

2024

Recommendation: Immediate policy intervention is required to restrict unauthorized housing societies.

Interactive Map

➤ Micro Level Hotspots Data (Selected City)

Multan Micro Lvel Zonel Analysis (Top Expansion Hotspots)

Area Name/ Location	Converted Area (sq km)	Total Impact (%)
ضلع ملتان, Gulshan, ملتان	4.42	0.63
تھمیل ملتان صدر, ضلع ملتان, Basti Bahawal Khara, ملتان	2.76	0.40
تھمیل ملتان صدر, ضلع ملتان, Lutfabad, ملتان	2.76	0.40

Download Full data (csv)

Download Boundary & Table Data (GeoJSON)

4. ERROR ANALYSIS & TROUBLESHOOTING LOG

The troubleshooting log reflects common engineering failure modes in satellite-data pipelines where spatial reference systems, tabular quality, rendering constraints, and data assumptions interact.

#	Real-World Issue	Root Cause	Resolution	Result
1	Validation points appeared offset on the map.	Input points and the rendered basemap were interpreted under different CRS assumptions.	Standardized the coordinate reference system before plotting and verified coordinate order.	Validation markers aligned spatially with the intended locations.
2	Streamlit map rendering became slow or timed out with larger spatial outputs.	Too many raw features were sent directly to the interactive layer without reducing payload size.	Reduced plotted geometry density, limited displayed records, and separated analytical data from visualization payloads.	Dashboard interaction became more stable and responsive.
3	Some CSV validation rows contained missing or malformed coordinate fields.	Manual point-entry records were not uniformly validated before ingestion.	Added row-level checks for missing latitude/longitude, numeric conversion, and valid geographic bounds.	Invalid records were isolated before spatial comparison.
4	NDVI/NDBI outputs contained unexpected extreme values.	Input spectral values included missing, inconsistent, or non-aligned band data.	Rechecked band inputs, normalized calculations, and added numerical sanity checks before visualization.	Index distributions became consistent with expected analytical ranges.
5	Classification results differed from manual observations in specific surface types.	Mixed pixels, ambiguous land boundaries, and spectral similarity caused localized misclassification.	Cross-checked mismatches against validation points and documented each case in the error-analysis artifact.	Errors became traceable and suitable for targeted refinement.

5. MULTI-LEVEL TEST CASES EVALUATION

Testing is organized from basic executable correctness to full-system integration. The progression ensures that computational accuracy is established before interactive rendering and cross-module behavior are judged.

Level	Complexity	Test Focus	Expected Outcome
Level 1	Low	Python / Streamlit syntax, imports, function execution, file paths, and application startup.	Notebook and Streamlit application execute without syntax or startup errors.
Level 2	Moderate	NDVI/NDBI calculation accuracy, valid numeric ranges, missing-value behavior, and band alignment.	Calculated indices match the implemented formulas and remain within expected analytical bounds after cleaning.

Level 3	Moderate-High	Dashboard interactivity, filters, controls, reruns, and Streamlit state management.	User interactions update relevant outputs without losing required application state or causing rendering failures.
Level 4	High	Validation-data mapping using >50 CSV rows, coordinate integrity, point-to-output matching, and mismatch handling.	All valid validation points map to the intended spatial context and mismatches are explicitly traceable.
Level 5	End-to-End	Complete Colab-to-Streamlit integration: processed data, classification outputs, validation evidence, and dashboard presentation.	The full workflow moves from processed satellite data to interactive UI outputs with reproducible evidence and documented error handling.

Testing maturity ladder



Acceptance principle: a passing visual result is not sufficient by itself. The final system must preserve numerical correctness, spatial integrity, traceable validation evidence, and stable interactive behavior across the complete processing chain.

6. CONCLUSION & DELIVERABLES STATUS

GeoAI: Satellite Data Processing & Land Classification System establishes an end-to-end technical workflow for converting satellite information into structured geospatial intelligence. The project combines quantitative spectral analysis, land classification, manual validation, error traceability, and interactive presentation within a cohesive architecture.

The completed implementation is production-oriented in its engineering structure: computation is separated from interface concerns, validation evidence is preserved in dedicated CSV artifacts, and spatial troubleshooting is treated as a formal quality-assurance activity. The result is a reusable foundation for future expansion into broader land-use classes, richer spatial analytics, larger datasets, or additional machine-learning models.

Deliverables Status

	Google Colab Notebook	GeoAI.ipynb
	Interactive Dashboard	app.py / Streamlit
	Validation Points CSV	50+ manual points
	Error Analysis CSV	15+ error cases
	Visualizations & Maps	interactive spatial outputs